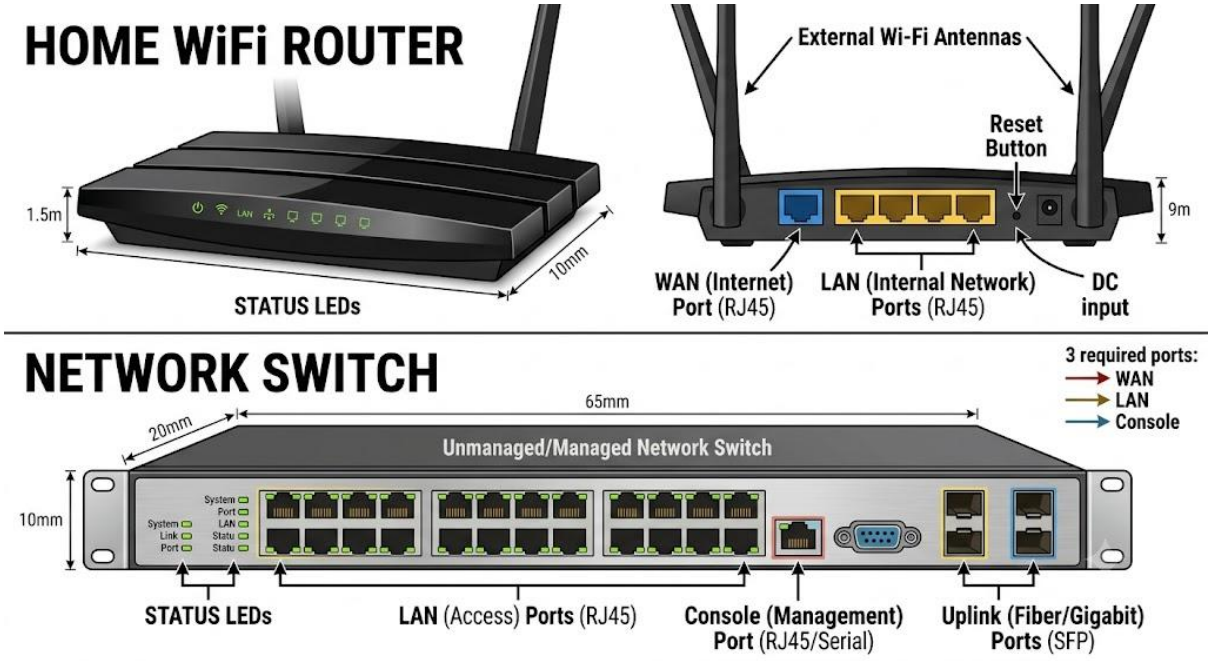
1.Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.



2.List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic.

Hint: Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

1. Network Scope & Connections

- **Router:** Connects different, distinct networks together (such as joining a private local area network to the global internet).
- **Switch:** Connects multiple local devices (computers, printers, servers) within the same single network (LAN).

2. Addressing & Data Units

- **Router:** Operates at Layer 3 (Network Layer) of the OSI model and uses **IP addresses** to forward data in the form of **packets**.
- **Switch:** Operates primarily at Layer 2 (Data Link Layer) and uses physical **MAC addresses** to forward data in the form of **frames**.

3. Traffic Management & Intelligence

- **Router:** Acts as a traffic director between networks, calculating the best path for data, managing network security (via firewalls/NAT), and assigning IP addresses (via DHCP).
- **Switch:** Filters and forwards traffic exclusively between specific ports on the local network by building a MAC address table, ensuring data goes directly to its targeted recipient device.

3.Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.



1. Home WiFi Router

A. Power Port (DC Input)

- Position: Located on the rear panel, typically next to the power button at the far end.
- Purpose: Receives direct current (DC) electricity from the external power adapter to power on the router hardware.

B. Ethernet Ports

- WAN Port (Internet):
    - Position: Located on the rear panel, color-coded in blue (or distinct from the other Ethernet ports).
    - Purpose: Connects via Ethernet cable directly to your ISP's modem or ONT (Optical Network Terminal) to bring the external internet connection into the device.

- LAN Ports (Local Area Network):
    - Position: A group of 4 contiguous RJ-45 ports on the rear panel, typically color-coded in yellow.
    - Purpose: Connect local wired devices (such as PCs, smart TVs, or network switches) directly to the router's local network.

C. LED Indicators

- Position: Located along the top front edge of the router facing forward for easy visibility.
- Purpose:
    - Power LED: Solid green indicates the device is powered on; amber/red indicates booting or hardware fault.
    - Internet/WAN LED: Indicates internet connectivity status and blinks when incoming/outgoing data packets pass through.
    - Wi-Fi LEDs (2.4GHz / 5GHz): Indicate that wireless broadcasting is active and flash during wireless data transmission.
    - LAN LEDs (1–4): Light up when a physical device is plugged into the respective LAN port and blink during active data transfer.

2. Network Switch

A. Power Port

- Position: Located on the rear or back corner of the switch body (a standard 3-pin AC power inlet or a round DC jack).
- Purpose: Provides electrical power required for the switch internal motherboard and port circuitry.

B. Ethernet Ports (RJ-45)

- Position: Arranged in numbered rows across the front panel (ranging from 5 to 48 ports depending on switch capacity).
- Purpose: Accepts standard RJ-45 Ethernet cables to connect multiple networked end-devices (computers, IP cameras, servers, or access points) together within the same local network (LAN).

C. LED Indicators

- System Status LEDs:
    - Position: Located on the front panel, usually near the left corner.
    - Purpose: Displays system health status (SYS, PWR, or STAT), indicating whether the switch is powered up and operating normally.
- Port Link/Activity LEDs:
    - Position: Built directly into the top corners of each individual RJ-45 Ethernet port.
    - Purpose:
        - Solid Light: Confirms an active physical connection (link established) to a device.
        - Blinking Light: Shows active data transmission/reception across that specific port.
        - Color (Green/Amber): Indicates connection speed (e.g., Green for 1000 Mbps Gigabit, Amber for 10/100 Mbps).

4.Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

To set up a fast, low-latency network for the gaming café, the ISP line (modem/ONT connection) connects directly to the **router** first.

Because the incoming line from the ISP delivers a single external public IP address, the router is required at the edge of the network to perform Network Address Translation (NAT) and manage security via an integrated firewall. The router assigns private local IP addresses to the café's devices using DHCP and directs traffic between the local network and the internet, allowing seamless streaming of live IPL matches.

To connect the 8 gaming PCs, an Ethernet cable runs from one of the router's LAN ports into an **8-port (or 16-port) Gigabit/Multi-Gigabit network switch**. Each of the 8 gaming PCs is then plugged directly into the switch using Cat6 Ethernet cables.

Justification for this setup:

- **Router First:** A standard switch cannot route traffic between different networks, run a firewall, or request/distribute IP addresses from an ISP on its own. The router must sit at the boundary to secure the network and route data between the internal café network and the outside world.
- **Switch for PC Connections:** Connecting all gaming PCs to a dedicated switch creates a high-speed Local Area Network (LAN). Switches process local traffic via MAC addresses at hardware speeds (Layer 2), ensuring zero-lag, dedicated bandwidth between PCs for local

multiplayer matches without putting unnecessary routing processing load on the router.